Aditya Sinha

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GitHub Profile
LinkedIn Profile
LeetCode Profile

Education

- **B Tech in Computer Science and Engineering (pursuing, 3rd year)** *2023–27*
  Vellore Institute of Technology, Vellore *(CGPA : 8.4)*
- **Senior Secondary (Class XII, TSBIE)** *2023*
  Sri Chaitanya Junior Kalasala, Hyderabad *(78.8%)*
- **Secondary School (Class X, CBSE)** *2021*
  Delhi Public School, Hyderabad *(94.6%)*

Personal Projects

- **VitalsFirst** [GH]
  An AI-driven patient triage and health monitoring system integrating machine learning with clinical decision support.
  - Built Random Forest model on clinical datasets with SHAP-based explainability and AUC-ROC evaluation.
  - Developed FastAPI backend serving ML predictions via REST APIs.
  - Designed Streamlit/React dashboard for clinician risk monitoring and analytics.
  - **Tech**: Python, FastAPI, Scikit-learn, SHAP, Streamlit, React, MySQL
- **DR Detection** [GH]
  Deep Learning-based Diabetic Retinopathy classifier
  - Trained CNN model for multi-class retinal image classification with preprocessing and augmentation.
  - Evaluated model performance using accuracy and validation metrics.
  - Integrated model into web interface for real-time prediction.
  - **Tech**: Python, TensorFlow/Keras, OpenCV, NumPy, Flask/Streamlit
- **Data Analytics Projects (Python)** [GH]
  Multi-domain EDA and predictive analysis
  - Built automated COVID-19 data pipeline with feature engineering and visualization.
  - Performed real-estate and global terrorism analysis using statistical and time-series techniques.
  - Developed trading risk notebooks analyzing drawdown and probability metrics.
  - **Tech**: Python, Pandas, NumPy, Matplotlib, SciPy, scikit-learn, Jupyter

Certifications

- **Python for Data Science, AI and Development (Coursera)**
- **Mastering Data Structures and Algorithms using C and C++ (Udemy)**
- **Ultimate Fusion 360 Course (Udemy)**
- **Gen AI Using IBM Watsonx (IBM)**

Technical Skills and Interests

Programming: C/C++, Java, Python, JavaScript, HTML/CSS
AI/ML Stack: Scikit-learn, OpenAI API, IBM Watsonx, FAISS, SHAP, NumPy, pandas, Matplotlib, SciPy, statsmodels
Frameworks & Tools: FastAPI, Streamlit, React.js, Node.js, VS Code, Git, GitHub, Jupyter,
Databases & Cloud: MySQL, MongoDB, Supabase, Google Colab, AWS
Core CS Subjects: DSA, OOP, DB, SE, CN
Strengths & Interests: AI and Web Application Development, Cloud Integration, ML Model Deployment, Data-Driven Problem Solving, Team Collaboration, Adaptability

Leadership and Extracurriculars

- **Core Committee Member – roboVITics** *2024 – 25*
- **Representative Student Ambassador – UMA (United Motorsport Academy)** *2023 – 25*
- **National Space Hackathon – IIT Delhi (Unstop)** *Apr – 25*